Worksheet - Can Use the Unit Circle (TU) - Solutions

Angles on the unit circle

Angle θ	**Quadrant?**	$\cos\theta$?	$\sin\theta$?
$\frac{\pi}{3}$	I	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$
$\frac{2\pi}{3}$	I	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$
$\frac{5\pi}{6}$	II	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$
$\frac{11\pi}{6}$	IV	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$
$\frac{3\pi}{4}$	I	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$
π	$-x$	-1	0
$\frac{7\pi}{4}$	IV	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$
$\frac{11\pi}{3}$	IV	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$
0	$+x$	1	0
$\frac{3\pi}{2}$	$-y$	0	-1
$\frac{31\pi}{6}$	I	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$
$\frac{11\pi}{4}$	IV	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$

Rotation practice

Problem 1

Show how to derive *all* these change of coordinate rules:

- Angle $\theta + \pi/2 \quad \gg\gg \quad (x,y) \to (-y, \quad x)$
- Angle $\theta - \pi/2 \quad \gg\gg \quad (x,y) \to (\quad y, -x)$
- Angle $\theta \pm \pi \quad \gg\gg \quad (x,y) \to (-x, -y)$

Solution:
Recall these essential identities from the “TD” learning target resources:

- $\cos(\theta + \pi/2) = -\sin\theta$
- $\sin(\theta + \pi/2) = +\cos\theta$

And these:

- $\cos(\theta + \pi) = -\cos\theta$
- $\sin(\theta + \pi) = -\sin\theta$

We can derive the remaining identities like this:

Derivations

$$\begin{aligned}
\cos(\theta - \pi/2) &\ggg \cos(-(-\theta + \pi/2)) \\
&\ggg \cos(-\theta + \pi/2) \\
&\ggg -\sin(-\theta) \\
&\ggg \sin\theta
\end{aligned}$$

$$\begin{aligned}
\sin(\theta - \pi/2) &\ggg \sin(-(-\theta + \pi/2)) \\
&\ggg -\sin(-\theta + \pi/2) \\
&\ggg -\cos(-\theta) \\
&\ggg -\cos\theta
\end{aligned}$$

$$\begin{aligned}
\cos(\theta - \pi) &\ggg \cos(\theta - \pi + 2\pi) \\
&\ggg \cos(\theta + \pi) \\
&\ggg -\cos\theta
\end{aligned}$$

$$\begin{aligned}
\sin(\theta - \pi) &\ggg \sin(\theta - \pi + 2\pi) \\
&\ggg \sin(\theta + \pi) \\
&\ggg -\sin\theta
\end{aligned}$$

By setting $x = \cos\theta$ and $y = \sin\theta$, all the rotation rules immediately follow.

Problem 2

If a point with coordinates (a, b) is on the unit circle at rotation θ, then the point on the unit circle at rotation $\theta + \frac{\pi}{2}$ has what coordinates?

Solution:
The formulas give $(x, y) \to (-y, x)$, so the answer is $(-b, a)$.

In case you forgot the formulas, we can derive this.

1. Set $(a, b) = (\cos\theta, \sin\theta)$.
2. Compute $\cos(\theta + \pi/2) \quad \ggg \quad -\sin\theta \quad \ggg \quad -b$.
3. Compute $\sin(\theta + \pi/2) \quad \ggg \quad \cos\theta \quad \ggg \quad a$.
4. Therefore the new point $(\cos(\theta + \pi/2), \sin(\theta + \pi/2))$ becomes $(-b, a)$.

Problem 3

If a point with coordinates (a, b) is on the unit circle at rotation θ, then the point on the unit circle at rotation $\theta - \pi$ has what coordinates?

Solution:
The formulas give $(x, y) \to (-x, -y)$, so the answer is $(-a, -b)$.

In case you forgot the formulas, we can derive this.

1. Set $(a, b) = (\cos\theta, \sin\theta)$.
2. Compute $\cos(\theta - \pi) \quad \ggg \quad -\cos\theta \quad \ggg \quad -a$.
3. Compute $\sin(\theta - \pi) \quad \ggg \quad -\sin\theta \quad \ggg \quad -b$.
4. Therefore the new point $(\cos(\theta - \pi), \sin(\theta - \pi))$ becomes $(-a, -b)$.

Problem 4

If a point with coordinates (x, y) is on the unit circle at rotation θ, and a point with coordinates (w, t) is on the unit circle at rotation $\theta - \frac{\pi}{2}$, then express w and t in terms of x and y.

Solution:

The formulas give $(x, y) \to (y, -x)$, so the answer is $w = y$ and $t = -x$.

In case you forgot the formulas, we can derive this.

1. Set $(x, y) = (\cos\theta, \sin\theta)$.
2. Compute:

$$\begin{aligned} \cos(\theta - \pi/2) \quad &\gg\gg \quad \cos(-(-\theta + \pi/2)) \\ &\gg\gg \quad \cos(-\theta + \pi/2) \\ &\gg\gg \quad -\sin(-\theta) \\ &\gg\gg \quad \sin\theta \\ &\gg\gg \quad y \end{aligned}$$

3. Compute:

$$\begin{aligned} \sin(\theta - \pi/2) \quad &\gg\gg \quad \sin(-(-\theta + \pi/2)) \\ &\gg\gg \quad -\sin(-\theta + \pi/2) \\ &\gg\gg \quad -\cos(-\theta) \\ &\gg\gg \quad -\cos\theta \\ &\gg\gg \quad -x \end{aligned}$$

4. Therefore the new point $(\cos(\theta + \pi/2), \sin(\theta + \pi/2))$ becomes $(-b, a)$.